Information Quality
Value and Business Impact

John R. Talburt, PhD, IQCP

Information Quality is About Relating Information to Value

- Maximizing the value of an organization’s information assets
- Making information products more valuable to the people who use those products

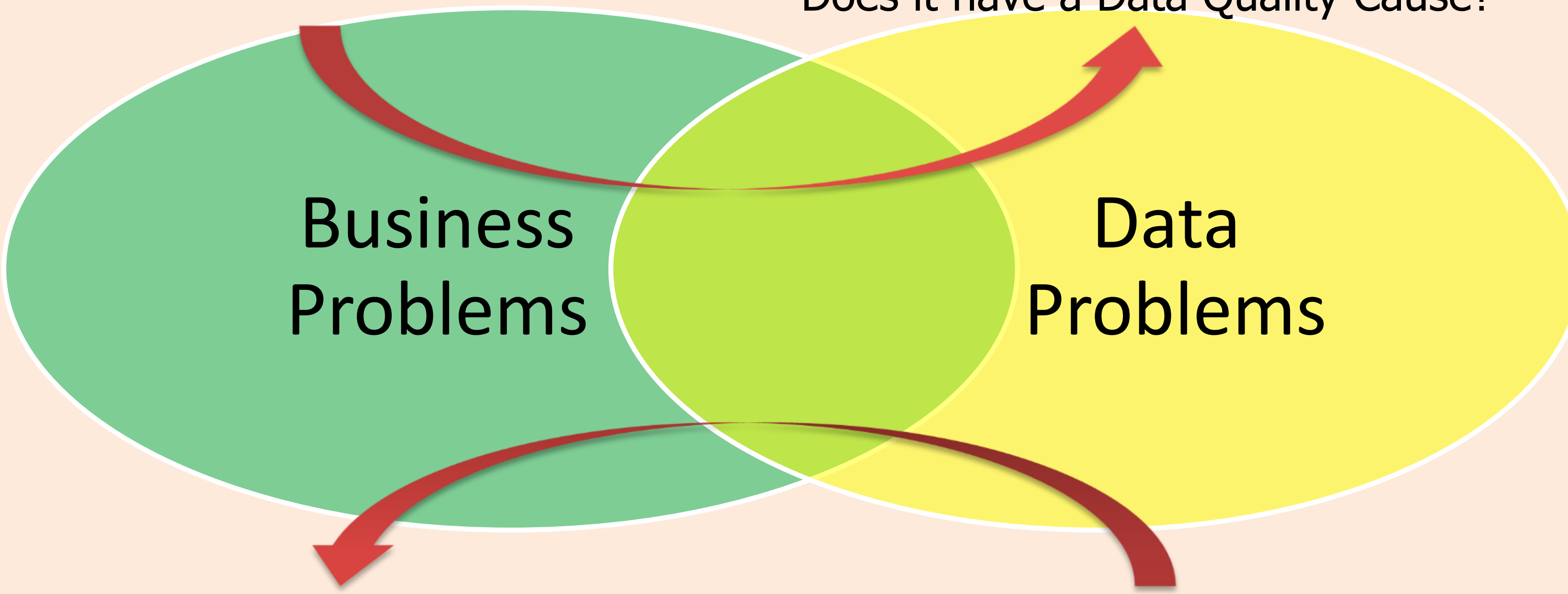
Two Ways to Relate Information to Value

Improve Business Results ← Improve Data Quality

↑

Start with a Business Issue

Does it have a Data Quality Cause?



Is it causing a business problem?

↓

Improve Data Quality

↓

Improve Business Results

Discover a Data Quality Issue

↑

Start with Data Quality Assessment

Places to Find Information Value

- Financial
  - Operating Costs
  - Revenue
  - Cash Flow
  - Charges
  - Budget
- Compliance
  - Government
  - Industry
  - Privacy
- Confidence
  - Trust
  - Decisions
  - Control
  - Forecasting
  - Reporting
- Satisfaction
  - Customer
  - Supplier
  - Employee

Places to Find Information Value

- Risk Mgmt
  - Regulatory
  - Credit
  - Investment
  - Competiveness
  - Leakage
- Productivity
  - Workloads
  - Throughput
  - Quality
- Health
  - Disease control
  - Treatments
  - Mental
  - Diet/Fitness
- Safety
  - Accident prevention
  - ….

CALCULATING
RETURN ON INVESTMENT (ROI)

Return on Investment (ROI)

- ROI is the net gain calculated as a percentage of the amount invested.
  - ROI = 100*(Net Gain) / (Cost of investment).
  - ROI should be positive
  - Good way to compare alternative opportunities
  - Should use “discounted” values for cost and revenue for long-term projects

Question

- If I can get a 5% annual return on my investment,
- What is today’s (present) value of a $100 payment to me one year from now?

Same question as

- How much money do I need to invest today to have $100 at the end of one year?

Solution

$$X + 0.05 \cdot X = 100$$

$$X \cdot (1 + 0.05) = 100$$

$$X = \frac{100}{1.05}$$

$$X = 95.24$$

Net Present Value of Money

- Basic assumption: Money should constantly be growing in value – annual discount rate X%
  - Un-invested money loses value
  - Future dollars are worth less today
- One million dollar jackpot prize
  - $100,000 paid today, then
  - $100,000 a year for next 9 years
  - At discount rate of 10%/yr, the prize is only worth **$675,902 today** **(Excel PV or NPV function)**

Excel Calculation

- Using NPV

  =100000+NPV(0.1,100000,100000,100000,
  100000,100000,100000,100000,100000,
  100000)

  = 675902.3816

- Using PV

  =-PV(0.1,10,100000,0,1)

  = 675902.3816

A Case Study Example (NFP)

- Not for profit company – problem with **inconsistent reference tables** across operating units – creating real cost in correcting erroneous shipments
  - Solution –set up **Single-Point-Of-Truth (SPOT)** corporate reference
- Analysis
  - Costs- Over 4 years = $120,000
  - Benefits- Over 4 years = $160,000
  - Simple ROI = (160-120)/120 = 33%

ROI Exercise

	Scenario 1		Scenario 2		Scenario 3	
Year	Cost	Benefit	Cost	Benefit	Cost	Benefit
1	**45,000**	**10,000**	**45,000**	**90,000**	**15,000**	**90,000**
2	**35,000**	**20,000**	**35,000**	**40,000**	**25,000**	**40,000**
3	**25,000**	**40,000**	**25,000**	**20,000**	**35,000**	**20,000**
4	**15,000**	**90,000**	**15,000**	**10,000**	**45,000**	**10,000**
Total	**120,000**	**160,000**	**120,000**	**160,000**	**120,000**	**160,000**

ROI Exercise

	Scenario 1		Scenario 2		Scenario 3	
Year	Cost	Benefit	Cost	Benefit	Cost	Benefit
1	**45,000**	**10,000**	**45,000**	**90,000**	**15,000**	**90,000**
2	**35,000**	**20,000**	**35,000**	**40,000**	**25,000**	**40,000**
3	**25,000**	**40,000**	**25,000**	**20,000**	**35,000**	**20,000**
4	**15,000**	**90,000**	**15,000**	**10,000**	**45,000**	**10,000**
Total	**120,000**	**160,000**	**120,000**	**160,000**	**120,000**	**160,000**
Simple ROI		**33%**		**33%**		**33%**
NPV						
NPV ROI						

Annual Discount Rate = 5%

ROI Exercise

	Scenario 1		Scenario 2		Scenario 3	
Year	Cost	Benefit	Cost	Benefit	Cost	Benefit
1	**45,000**	**10,000**	**45,000**	**90,000**	**15,000**	**90,000**
2	**35,000**	**20,000**	**35,000**	**40,000**	**25,000**	**40,000**
3	**25,000**	**40,000**	**25,000**	**20,000**	**35,000**	**20,000**
4	**15,000**	**90,000**	**15,000**	**10,000**	**45,000**	**10,000**
Total	**120,000**	**160,000**	**120,000**	**160,000**	**120,000**	**160,000**
Simple ROI		**33%**		**33%**		**33%**
NPV	**108,540**	**136,261**	**108,540**	**147,499**	**104,217**	**147,499**
NPV ROI		**26%**		**36%**		**40%**

Annual Discount Rate = 5%

MAKING THE BUSINESS CASE FOR INFORMATION QUALITY

The Importance of a Business Case

- You must develop a convincing business case for IQ improvement to be successful
- A sound business case will
  - Help management see IQ improvement as an "investment" not an added expense
  - Gain support from management
  - Justify the resources necessary to have a successful project

Align with Business Goals

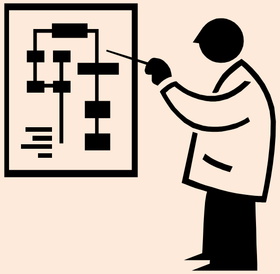
The Business Case must make a clear connection between

- Information quality improvement
- Achieving business goals and objectives

Understand the Full IQ Context

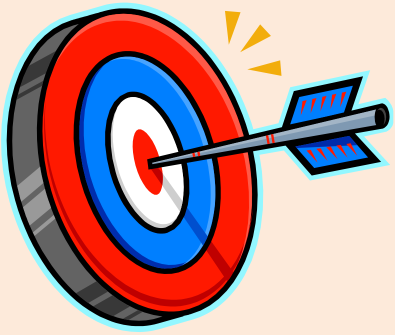
Understand the

- Data, what's in it, what data items are most important
  - Critical Data Element (CDE) analysis
  - What would be the impact if missing or inaccurate
- People using the data, what departments
- Processes transforming the data
- Technology/software being using



May be Many Targets to Improve

- For each opportunity, estimate the Value = Benefits – Costs
- Benefit may mean
  - Increased revenue
  - Lower operational cost
  - Avoiding fines and penalties
- Cost may mean
  - Additional software
  - Additional personnel



Weigh the Alternatives

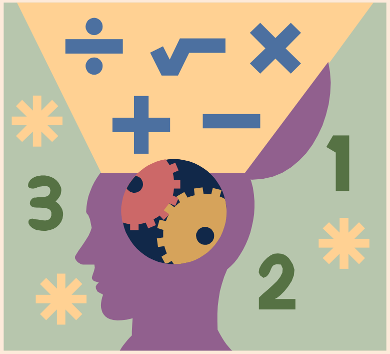
- Map the opportunities into a cost-benefit matrix

BENEFIT ↑ / COST →		
	High Benefit Low Cost, High Value 1st Choice	High Benefit High Cost, Medium Value 2nd Choice
	Low Benefit Low Cost, Medium Value 3rd Choice	Low Benefit High Cost, Low Value 4th Choice

BENEFIT

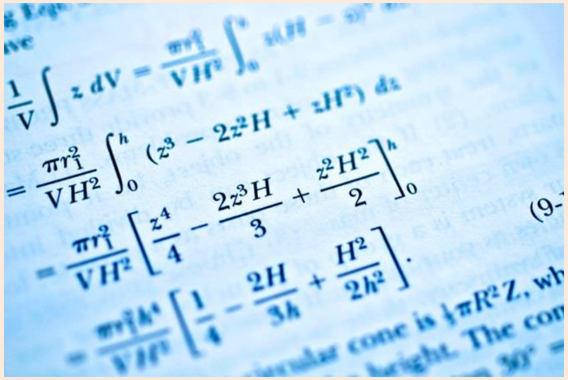
COST

Write the Business Case



- Qualitative measures are useful, but always include a **quantitative financial justification**
- Clearly explain how the benefits of the project will be greater than the investment
  - i.e. the project will create value
- The business case should **"sell" the project** to management – It should be something that they want to buy

Be Quantitative



Example Benefit

- Qualitative: Project A to improve information product X will increase customer satisfaction
- Quantitative: Project A is expected to increase customer retention by 10% a year. Cost to acquire new customer is $1,200, Cost to retain a customer is only $400. For 2,000 customers Savings = 0.1*2,000*(1,200-400) = $160,000/yr

Consider All the Factors

In addition to analyzing benefit and cost

- How much time will it take?
- Will it have a long-term benefit?
- What is the risk of failure?
- Are there enough people in the right roles to carry out the project – IT, Data Experts, Project Leaders, etc.
- How visible will the project be? The results?

CHARACTERISTICS OF INFORMATION

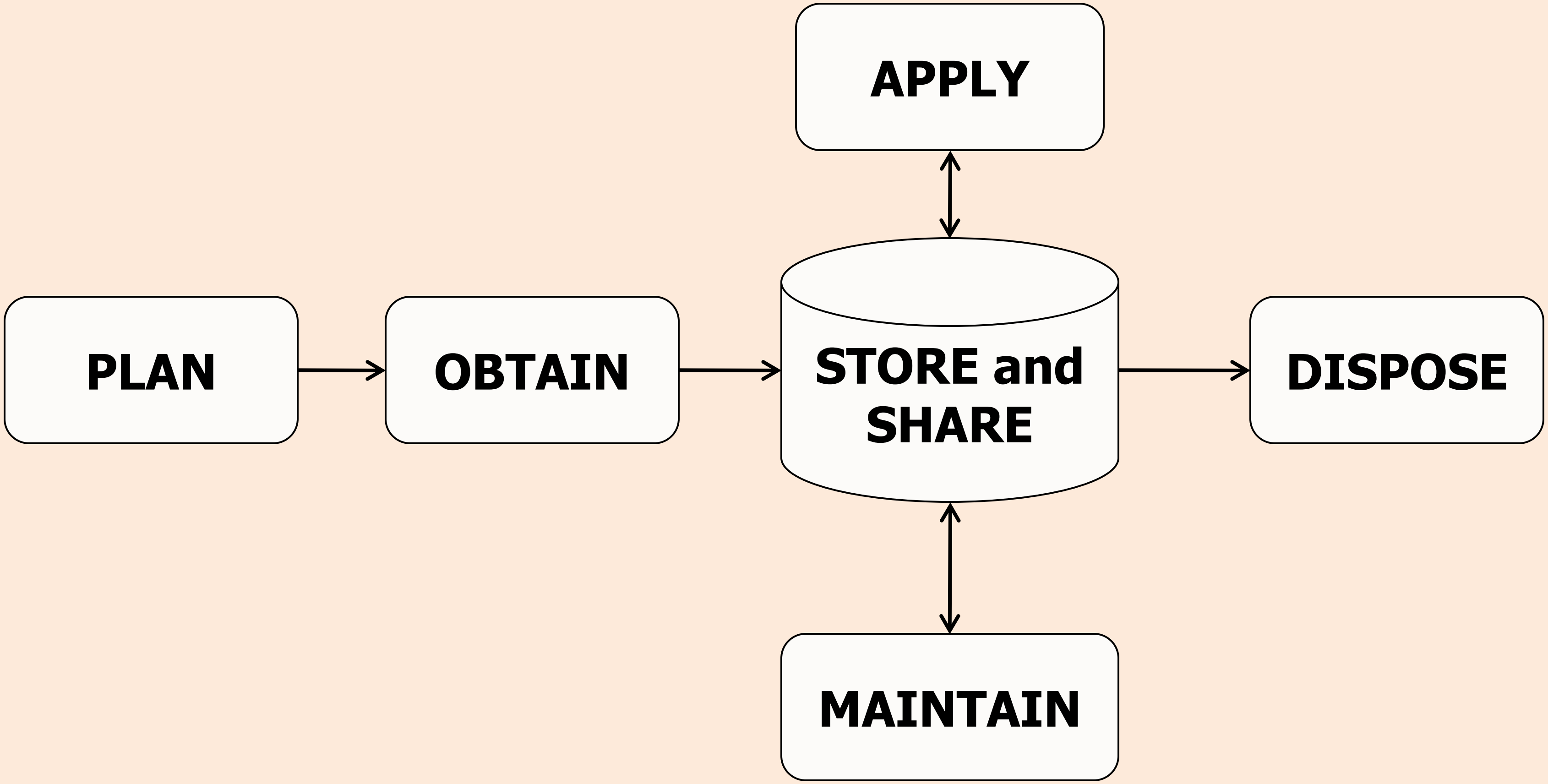
The Information Life Cycle

- **Plan** for information
- **Obtain** the information
- **Store and Share** the information
- **Maintain** and manage information
- **Apply** the information to accomplish your goals
- **Dispose** of the information it is no longer needed

POSMAD*

*From Danette McGilvray, *Exectuing Data Quality Projects*

Life-Cycle is not a Linear Process



Plan Life-Cycle Activities

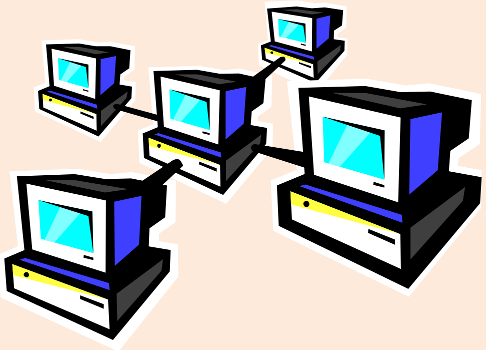
- Plan (Prepare for the resource)
  - Identify objectives, plan information architecture, develop standards and definitions.
  - When modeling, designing, and developing applications, databases, processes, organizations, etc., many activities should be considered part of the Plan phase for information
  - Understand limitation and restrictions on the use of the data – regulatory, privacy

Obtain Life-Cycle Activities

- Obtain (Acquire the resource)
  - Create or compile records
  - Purchase/rent data
  - Load external files
  - Extraction and capture of unstructured information
    - Text
    - Image
    - Audio

Store and Share Life-Cycle Activities

- Store and Share
  - Store data electronically in databases or as some type of file, or store hardcopy as a paper or image
  - Share physical access to the information through networks, web pages, blogs, wikis, an enterprise service bus, or email
  - Share logical access to the information through data catalogs and metadata specifications



Maintain Life-Cycle Activities

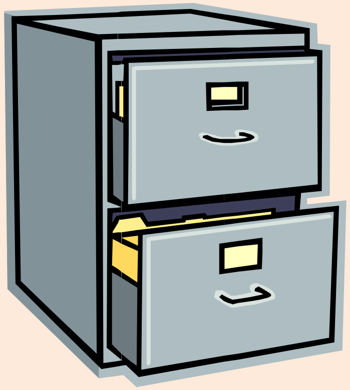
- Maintain (Ensure that the resource continues to work properly)
  - Update, change, manipulate, parse, standardize, validate, or verify data.
  - Enhance or augment data cleanse, scrub, or transform data.
  - De-duplicate, link, or match records.
  - Merge or consolidate records

Apply Life-Cycle Activities

- Apply (Use the resource to accomplish your goals and objectives)
  - Retrieve and use information
  - Create information products for users
    - Completing a transaction
    - Writing a report
    - Making a management decision reports
    - Running automated processes

Dispose Life-Cycle Activities

- Dispose (Move information off-line when direct access is no longer needed)
  - Archive information for later access
  - Delete or destroy data or records for legal reasons

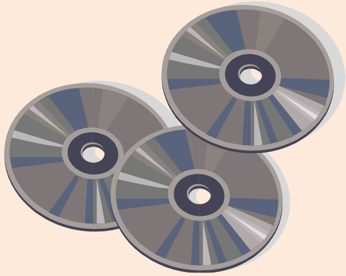


Life-Cycle Value, Cost, and Quality

- All six phases of the Information Life Cycle carry cost
- Only in the APPLY Phase does the organization and the user receive VALUE from information – the other five phases represent only overhead!
- IQ can be affected by activities in all of the phases in the life cycle.

Information is a Unique Resource

- As a raw material, data is not consumed when it is used to make information products,
- Data and information can be easily copied and used (and sold) over and over again with almost no additional cost
  - Quality is critical – copies of bad data are bad data
  - The marginal value increases the more it is used



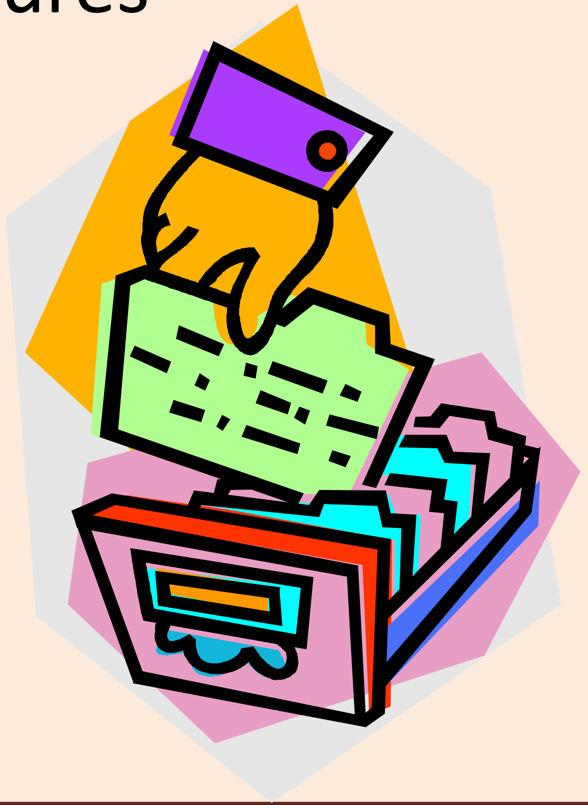
Information is a Non-Fungible Asset

- Fungible assets are interchangeable assets
  - One barrel of oil is just like another barrel of oil
  - One concrete block is the same as any other
- Information describes different identifiable entities
  - Customers
  - Products
  - Services
  - That are not interchangeable

UNDERSTANDING DATA CATEGORIES

Data Categories

- Data categories are groupings of data with common characteristics or features
- They are useful for managing the data because certain data may be treated differently based on their classification



Data Categories By Structure

- Structured Data
- Unstructured Data
- Semi-Structured Data

Structured Data

- Directly machine readable
- Has a formal syntax,
  - Relational Databases
  - CSV files
  - XML Documents
  - JSON Documents

Unstructured Data

- Human readable
- Require pre-processing to be machine readable
  - Text: reports, social media
  - Images: photos, videos
  - Audio: audio transcripts, video soundtracks

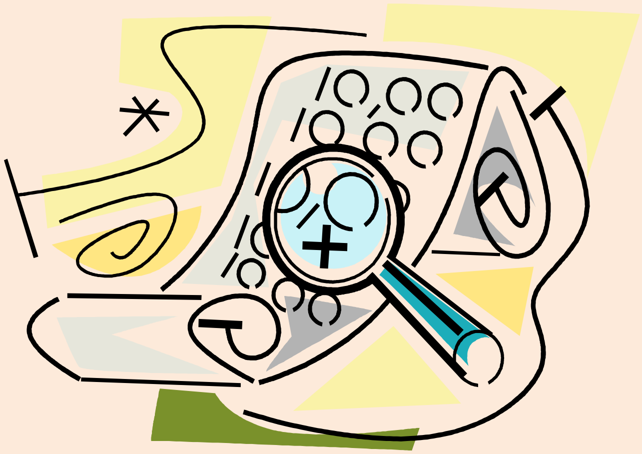
Semi-Structured Data

- Part Structured – Part Unstructured
- To some extent almost all data are semi-structured – a matter of degree
- Examples
  - Comments column in a database table
  - Free-form address in a database table
  - E-mail records – easy to machine read To:, From:, Date:, Subject:, but not meaning of message

Data Categories by Function

- Operational Data:
  - Master Data
  - Transactional Data
  - Reference Data
  - Metadata
- User Generated Data
- Machine Generated Data

Master Data



- Master data describe people, places, and things critical to the organization
  “Nouns of the data vocabulary”
- Each master entity is assigned a unique and persistent identifier (e.g. UALR T-Number)
- Identifiers are appended to incoming records
- Simplifies the deduplication and integration of records for downstream business units

Transactional Data

- Describe events that take place as an organization conducts its business "Verbs of the data vocabulary"
  - Sales orders
  - Credit card payments
  - Insurance claims

SalesAssoc: S123 → SOLD Product: P456 → TO CustID: C789

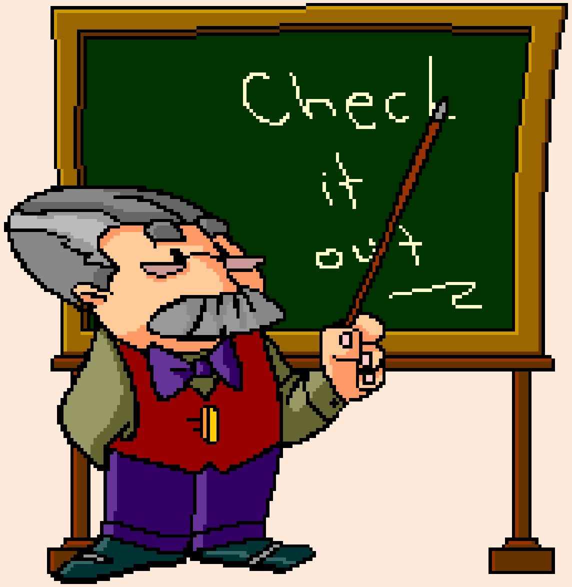
Reference Data

- Sets of valid codes or classification schemes that are referenced by
  - Transaction records
  - Master records
  - Applications

- Key to data integration and data interoperability
  - External: Postal codes, UPC
  - Internal: Job codes, building codes

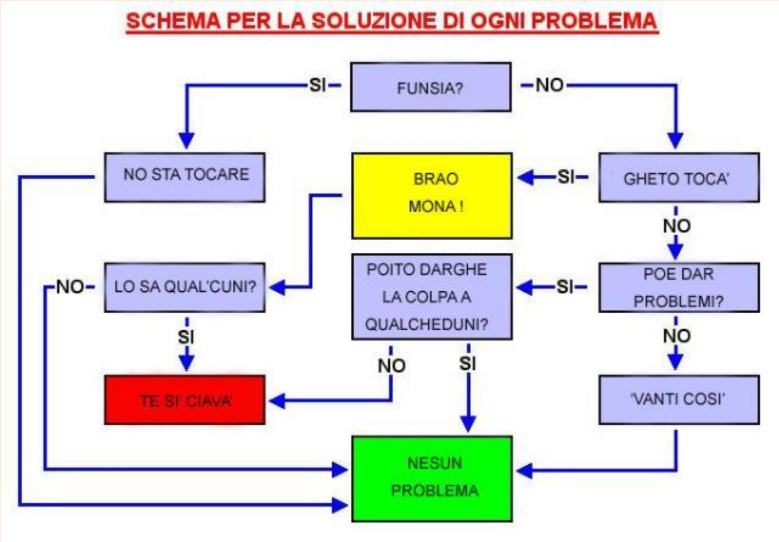
Metadata

- Metadata literally means "data about data."
- Metadata label, describe, or characterize other data and make it easier to retrieve, interpret, or use information.
  - Technical metadata
  - Business metadata



Types of Metadata

- Technical metadata describe technology, systems, data structure, data quality rules, and data standards
- Business metadata define business terminology and business rules
- Process metadata define processing lineage



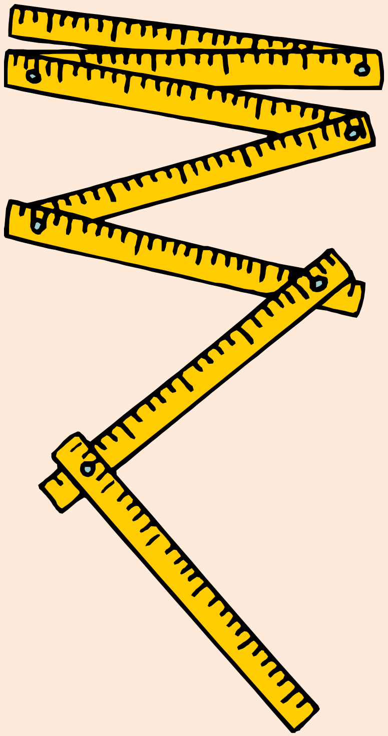
Metadata Management and Tools

- Data
  - Data Catalogs – Track physical content/location, profile statistics
  - Business Glossary – Document logical content, data quality requirements
- Processes
  - Process metadata document and run IT processes
  - Business Process Management tools automate and regulate business processes

DATA QUALITY AND DATA REQUIREMENTS

Data Requirements

- Data requirements include
  - Data models and schemas
  - Business rules
  - Data quality requirements
  - Data standards



Data Specifications

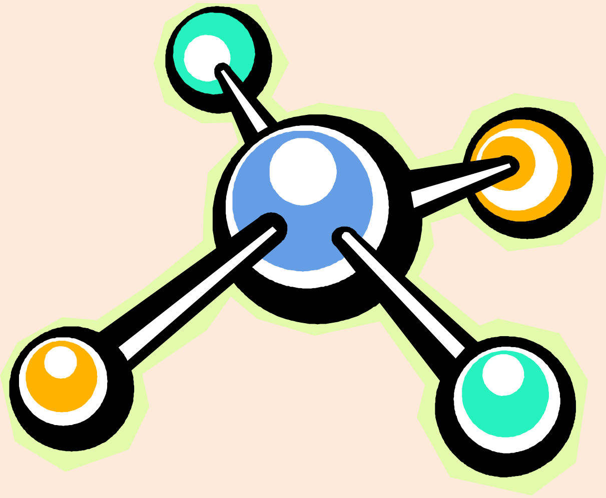
- **Data Quality is conformance to Data Requirements**
- Missing or poorly defined data requirements lead to poor data quality!

Data Standards

- Data standard is a collection of related data requirements to ensure everyone in an organization is acting in a consistent way
  - Name data
  - Define it (metadata)
  - Format it
  - Establish valid values
  - Specify business rules

Data Models

- A way of visually representing the structure of an organization's data.
- It is also a specification of how data are to be represented in a data store
- Data model view of the data
  - Conceptual
  - Logical
  - Physical



Questions and Discussion